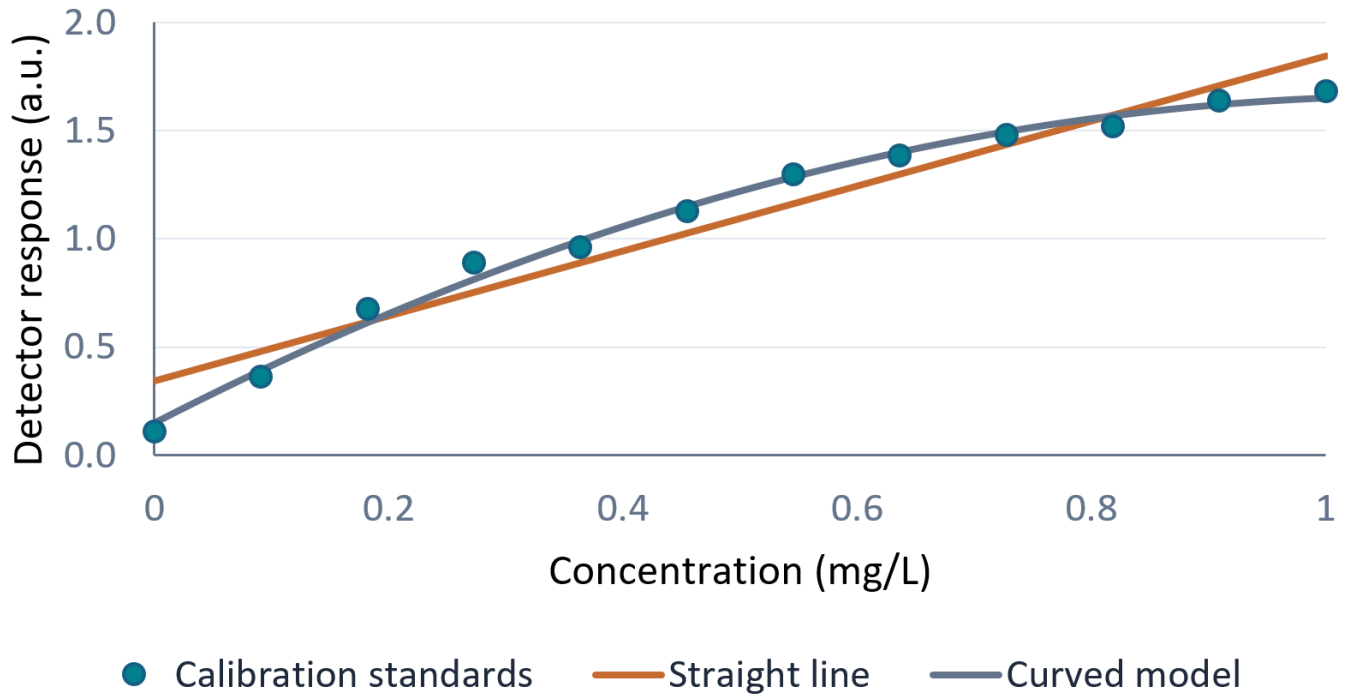
Multiple failure modes that can lead to underperforming models

Failure mode	What goes wrong?	Useful check
Underfitting	The model misses a useful pattern.	Training and validation performance
Overfitting	The model fits accidental details.	The training–validation gap
Bias in the data	Some conditions have poor coverage.	Sampling and performance by group
Out of distribution	New conditions differ from those studied.	The intended conditions of use
Information leakage	Training and test information overlap.	Split units and data preparation

Underfitting is biased towards assumptions of the underlying (simple) trend of the data

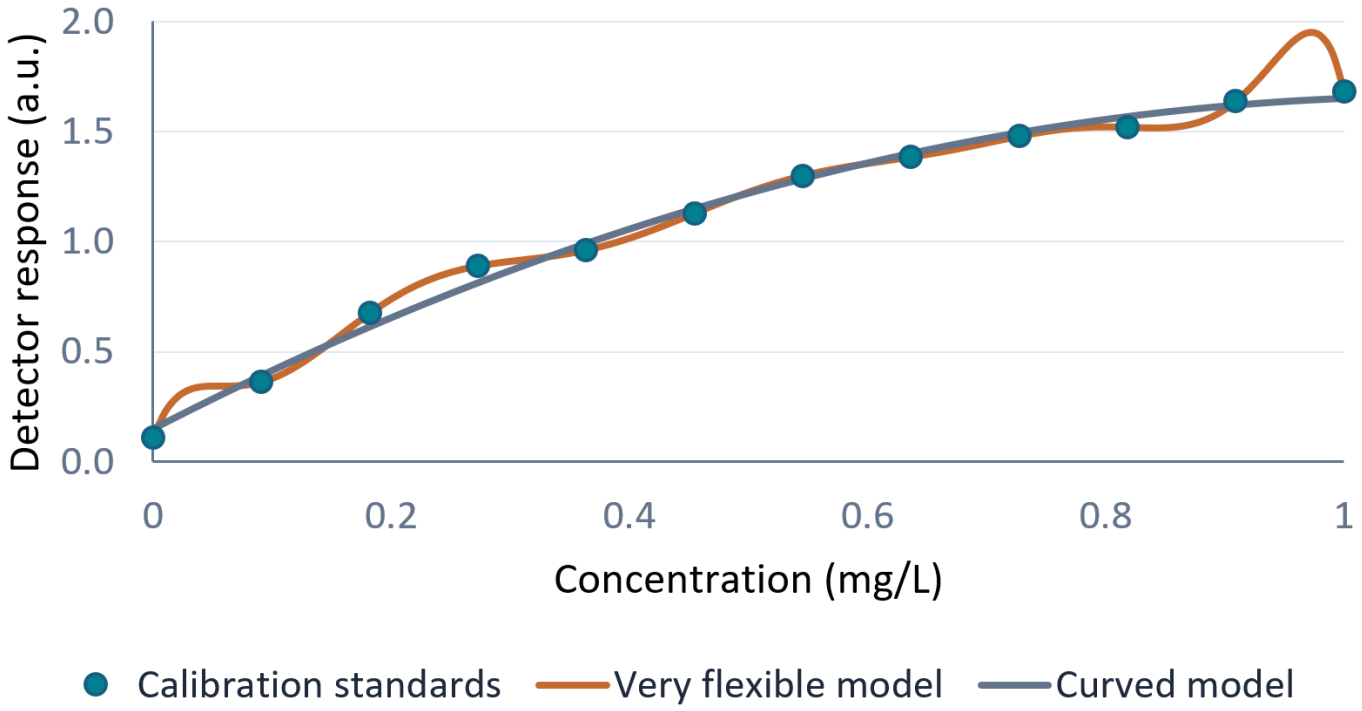


A straight line misses the bend as the detector approaches saturation.

Training and validation errors both exceed those of a suitable curved model.

A more suitable model can describe this pattern.

Overfitting fits too much variance between the features and targets

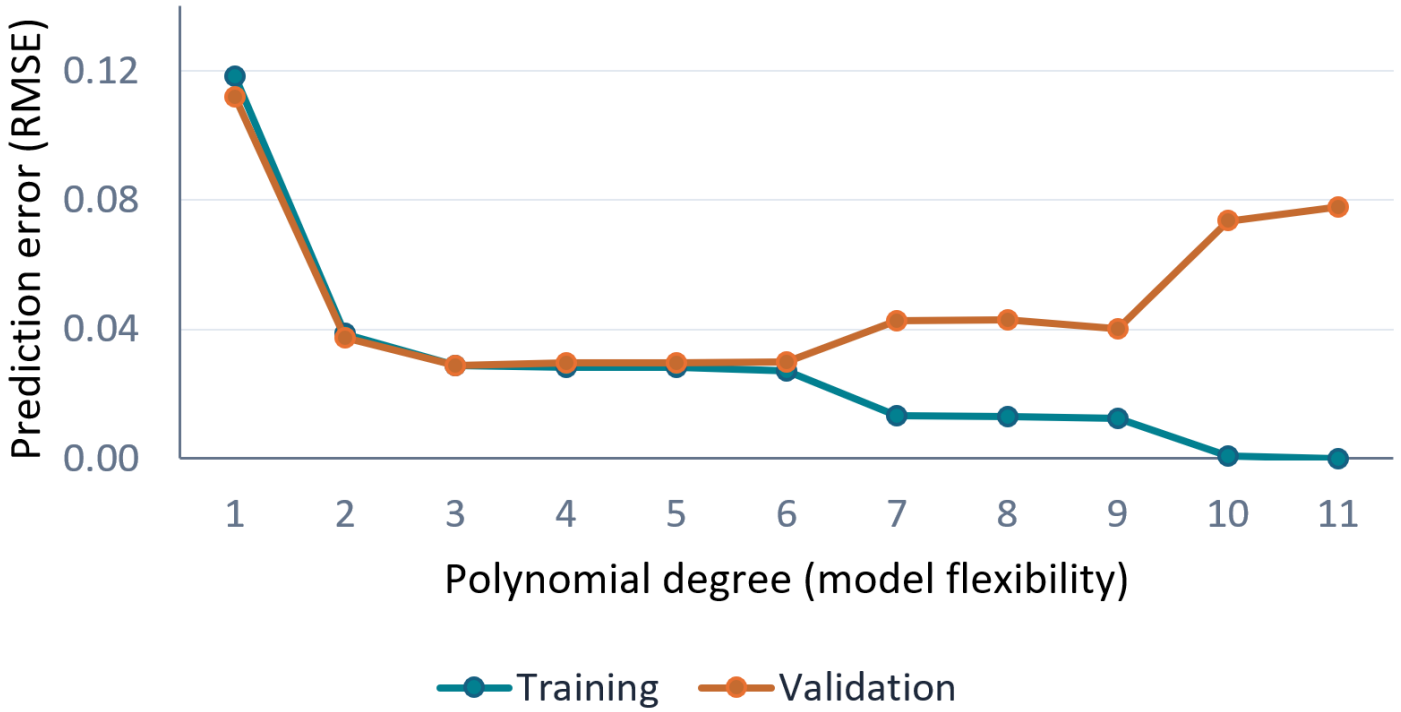


The flexible curve passes through every training standard.

Extra bends follow noise and increase error on new validation measurements.

Perfect training agreement can give poorer new predictions.

Typical trend for model performance when increasing model complexity

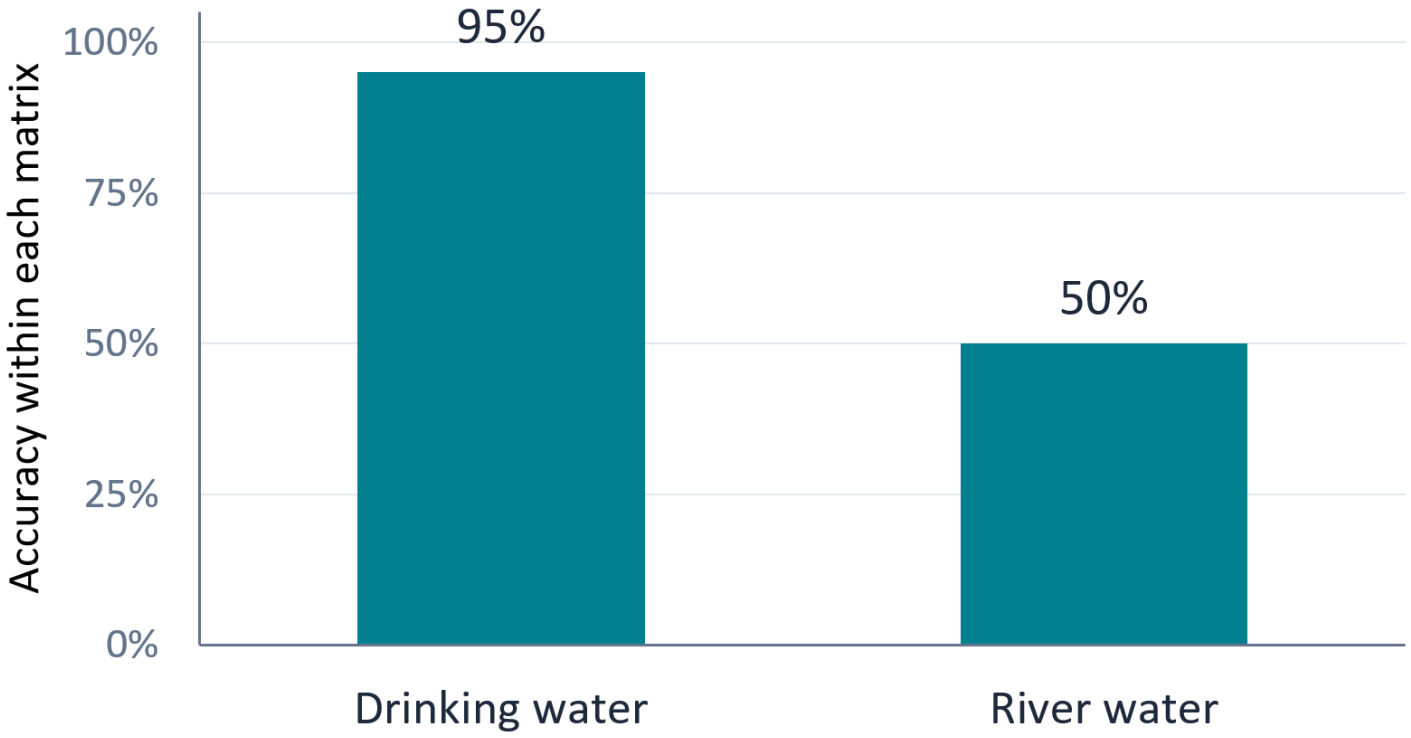


More flexibility keeps reducing training error.

Validation error falls, then rises as the model follows measurement noise.

Choose with validation. A separate test checks the fixed choice.

Bias in training data can affect a model negatively

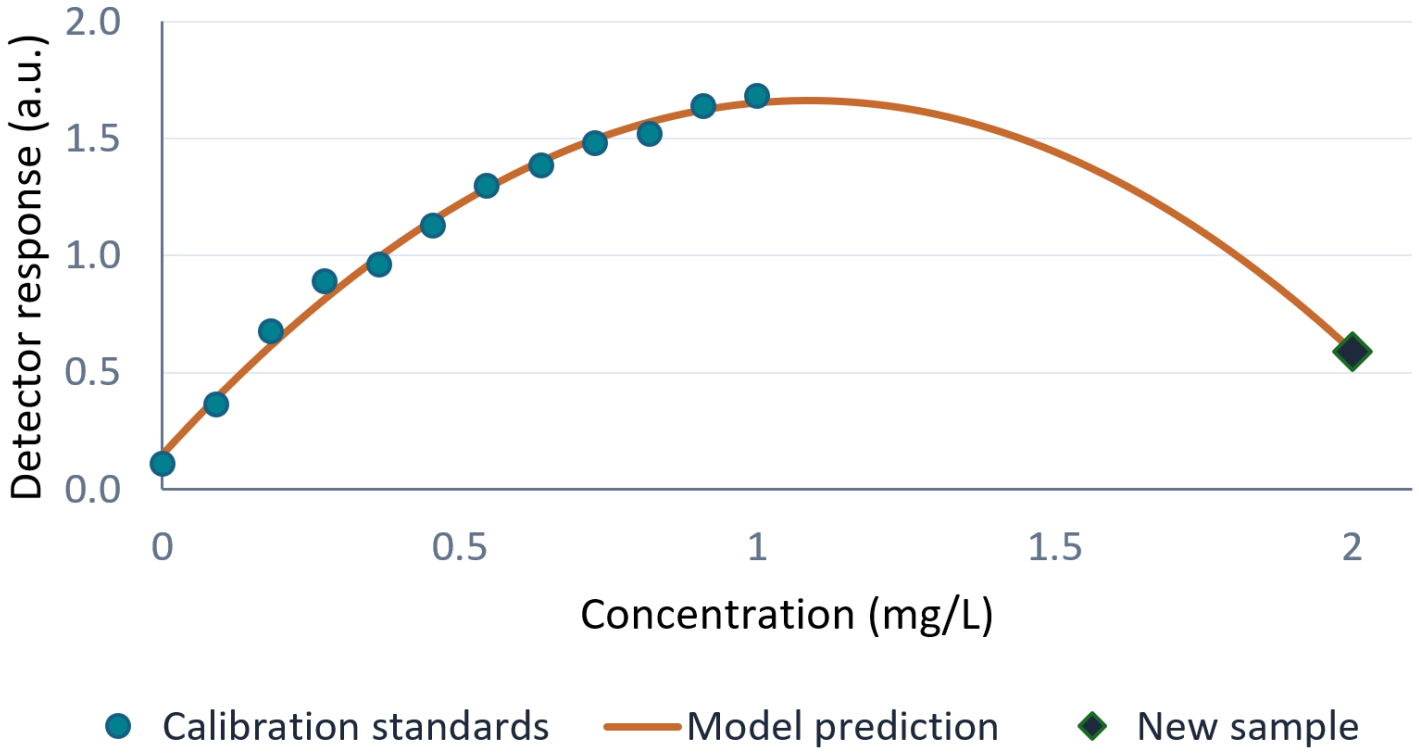


A contaminant classifier learned mostly from drinking-water samples.

Hypothetical test set:
900 drinking-water samples
100 river-water samples

Overall accuracy is 90.5%, although river-water accuracy is only 50%.

Interpolation is easy, extrapolation is difficult; variety in training data!

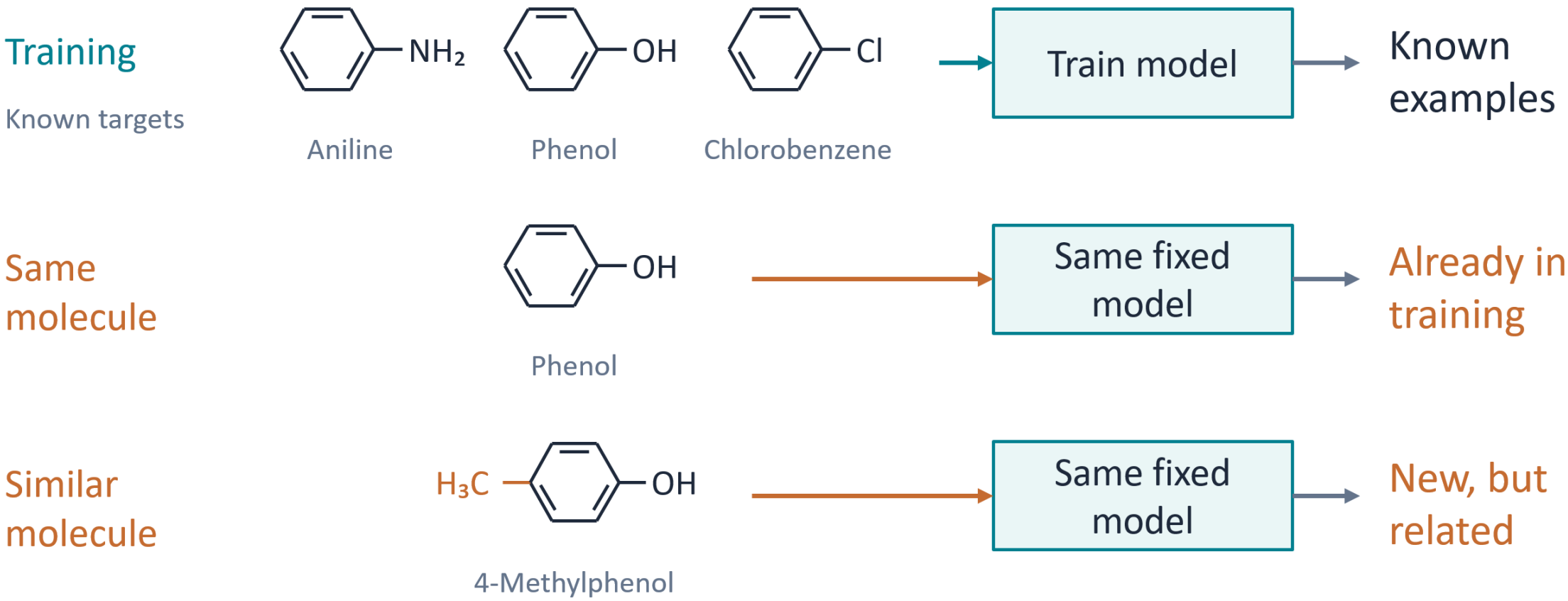


Standards cover 0–1 mg/L. The new sample is at 2 mg/L.

Predictions at 2 mg/L have no validation measurements to support them.

New instruments or sample matrices can also change the conditions of use.

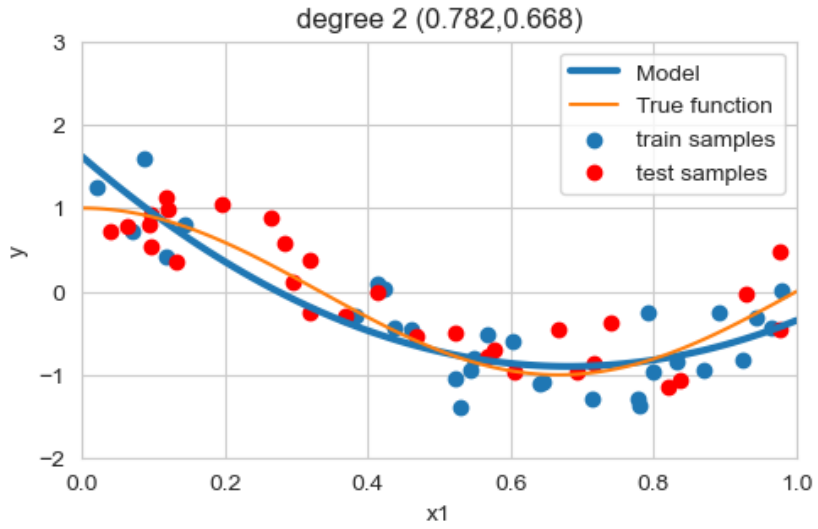
Estimating performance is difficult; you want variety and differences, but not too much extrapolation

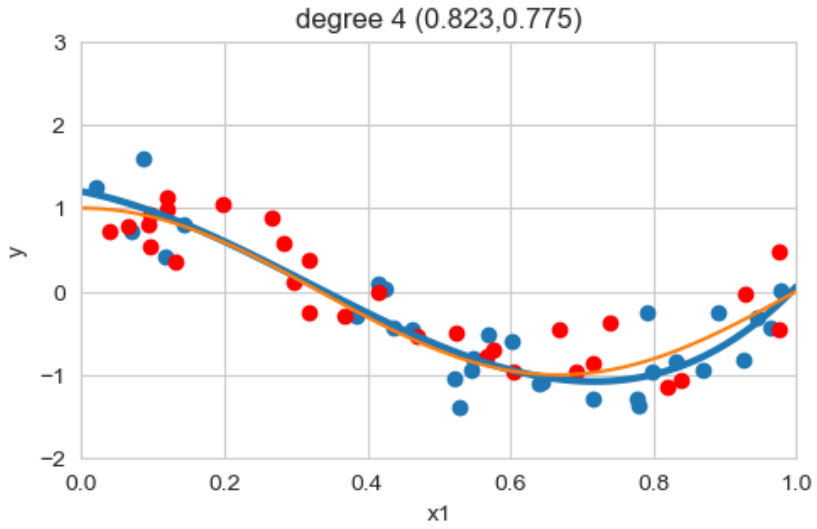


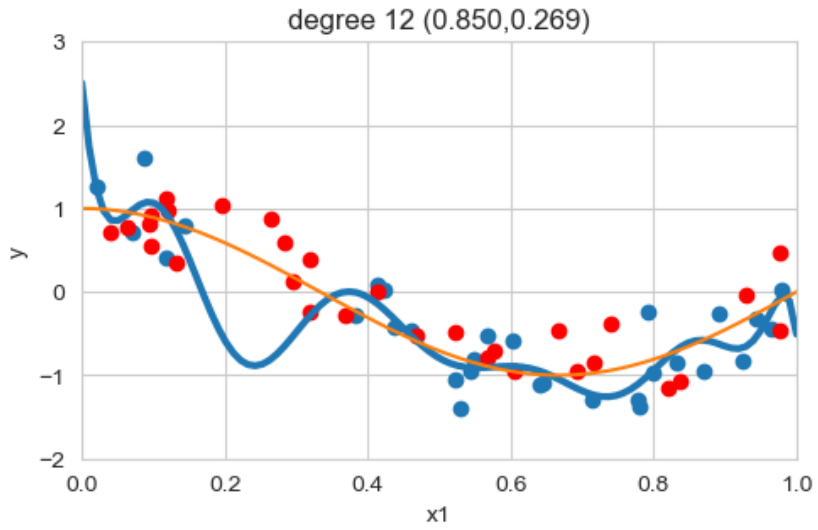
Regression models the behavior between features and continuous target values

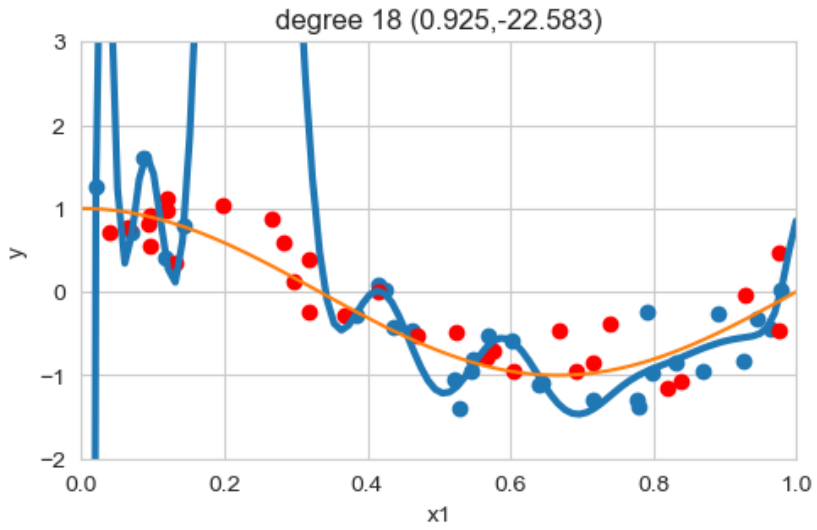
Failure mode	What goes wrong?	Useful check
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Overfitting with difference degrees of polynomials









label noise

feature noise

feature relevance

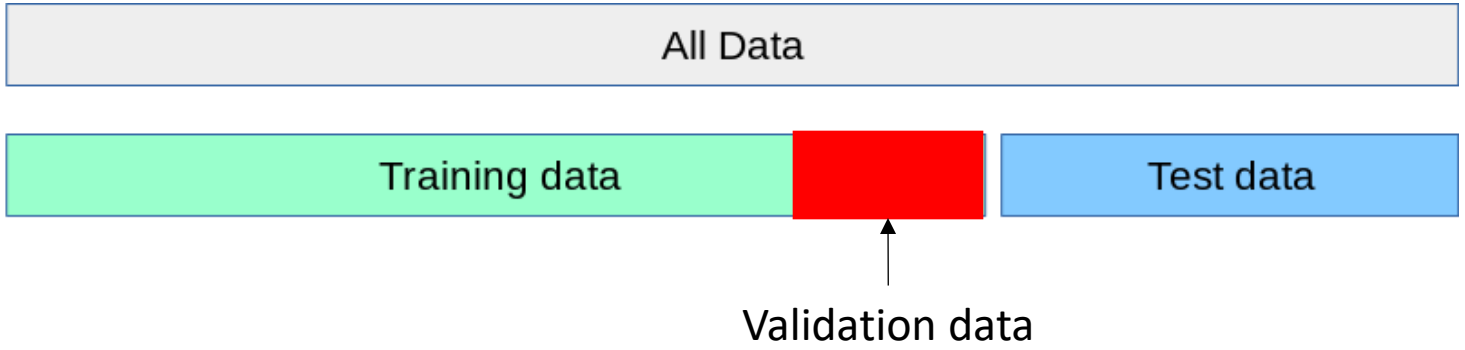
relatively small train sets

don’t try to fit the data perfectly

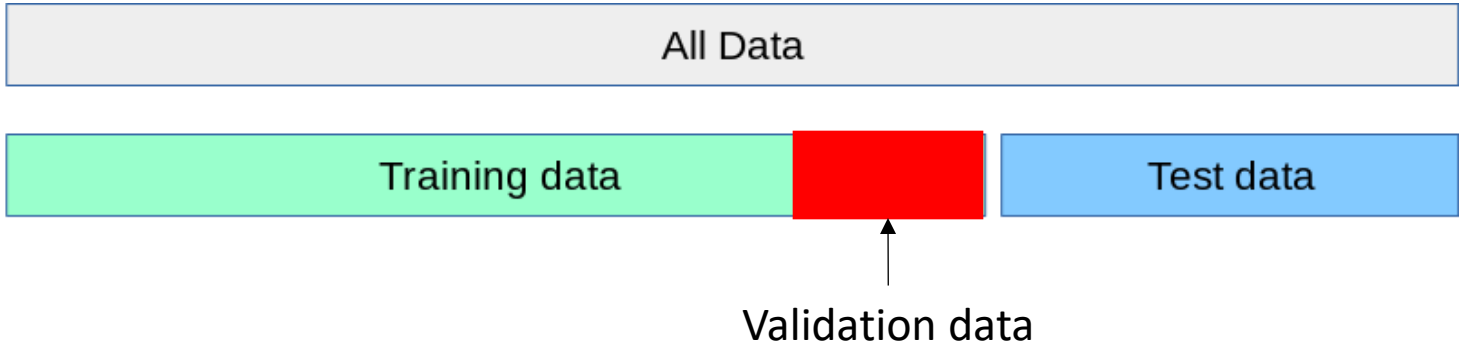
don’t try to use all features

notice how R^2 on the train set does increase with *d*

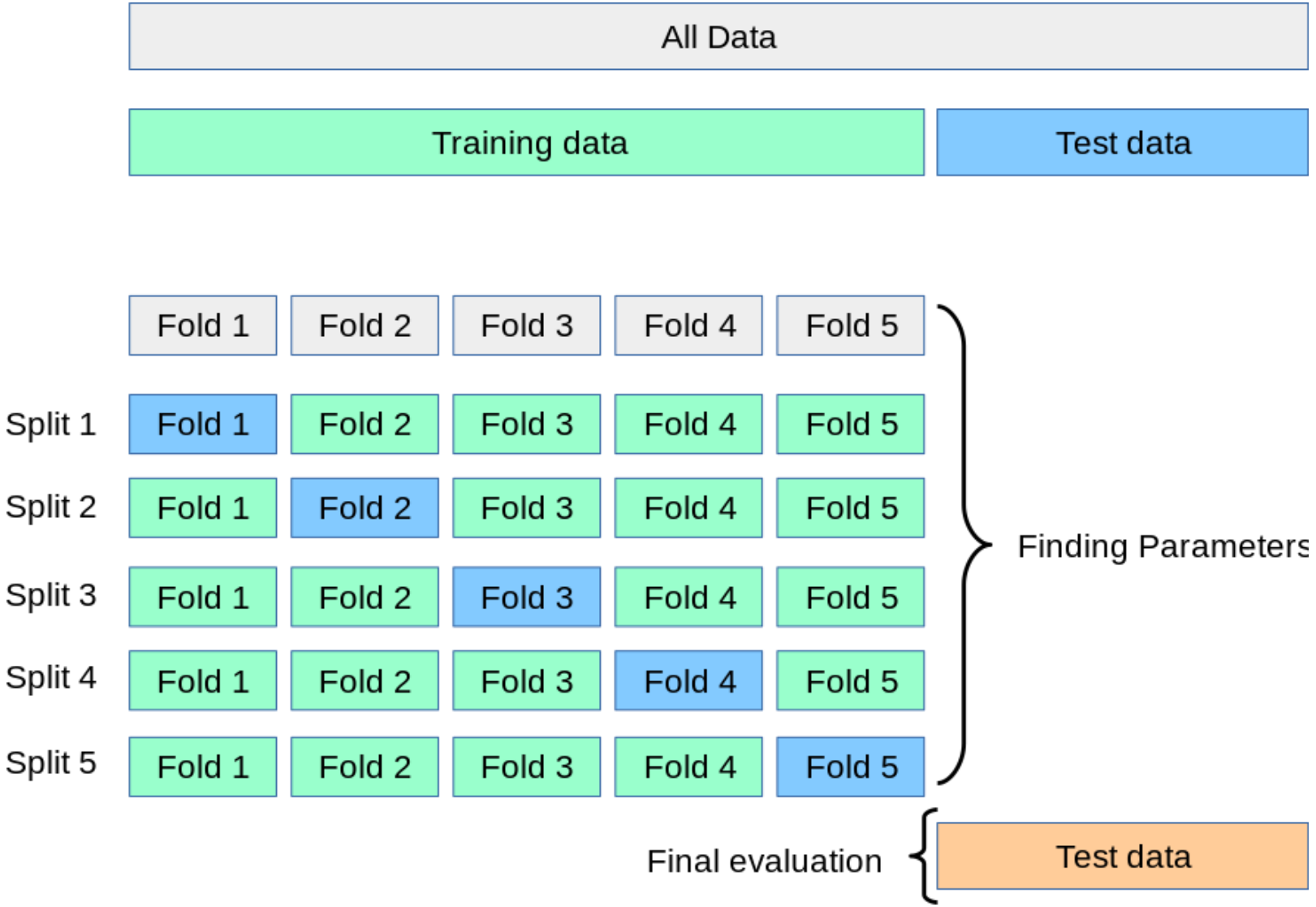
Validation hold-out set



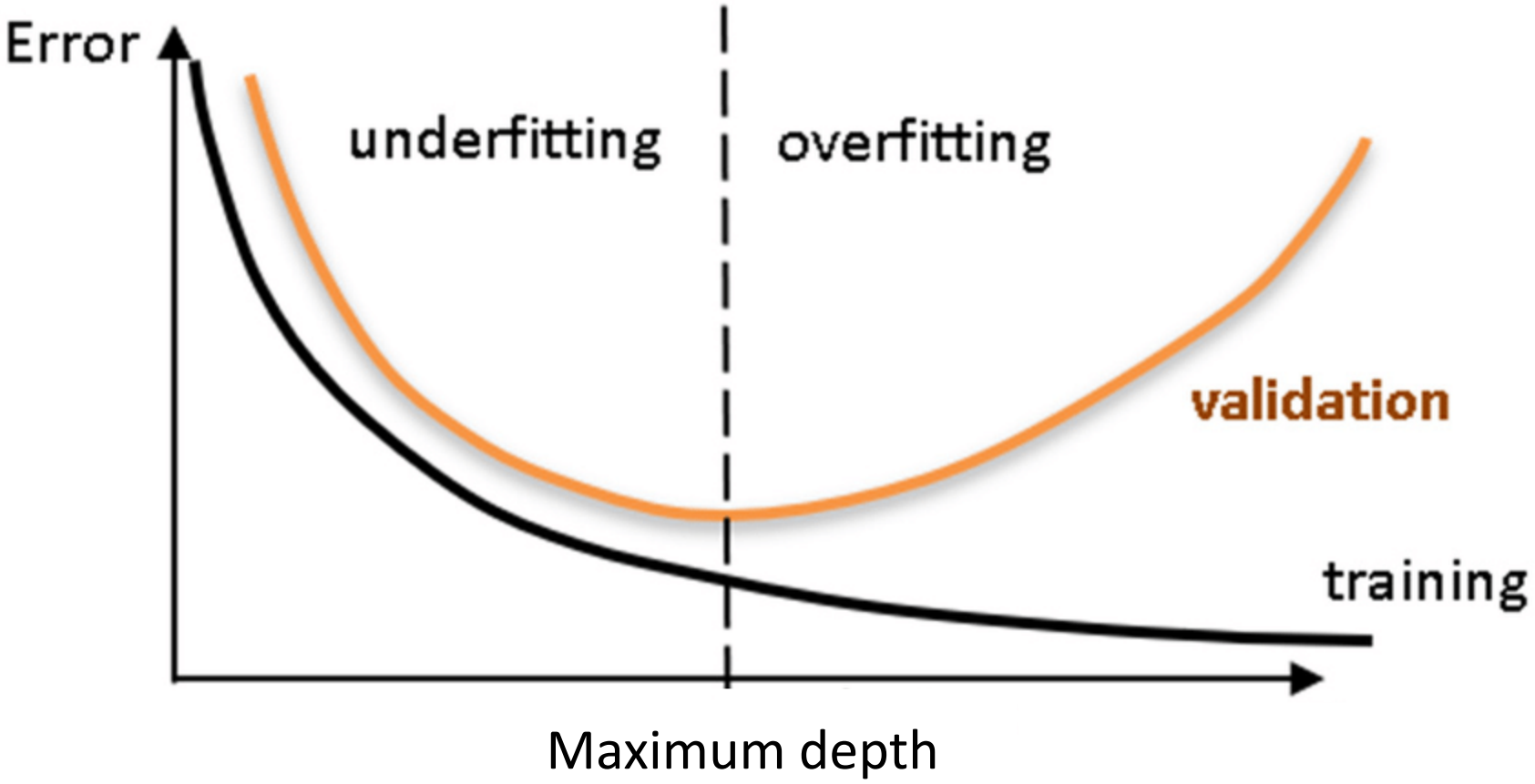
Validation hold-out set



5-fold cross-validation



Tuning regularization

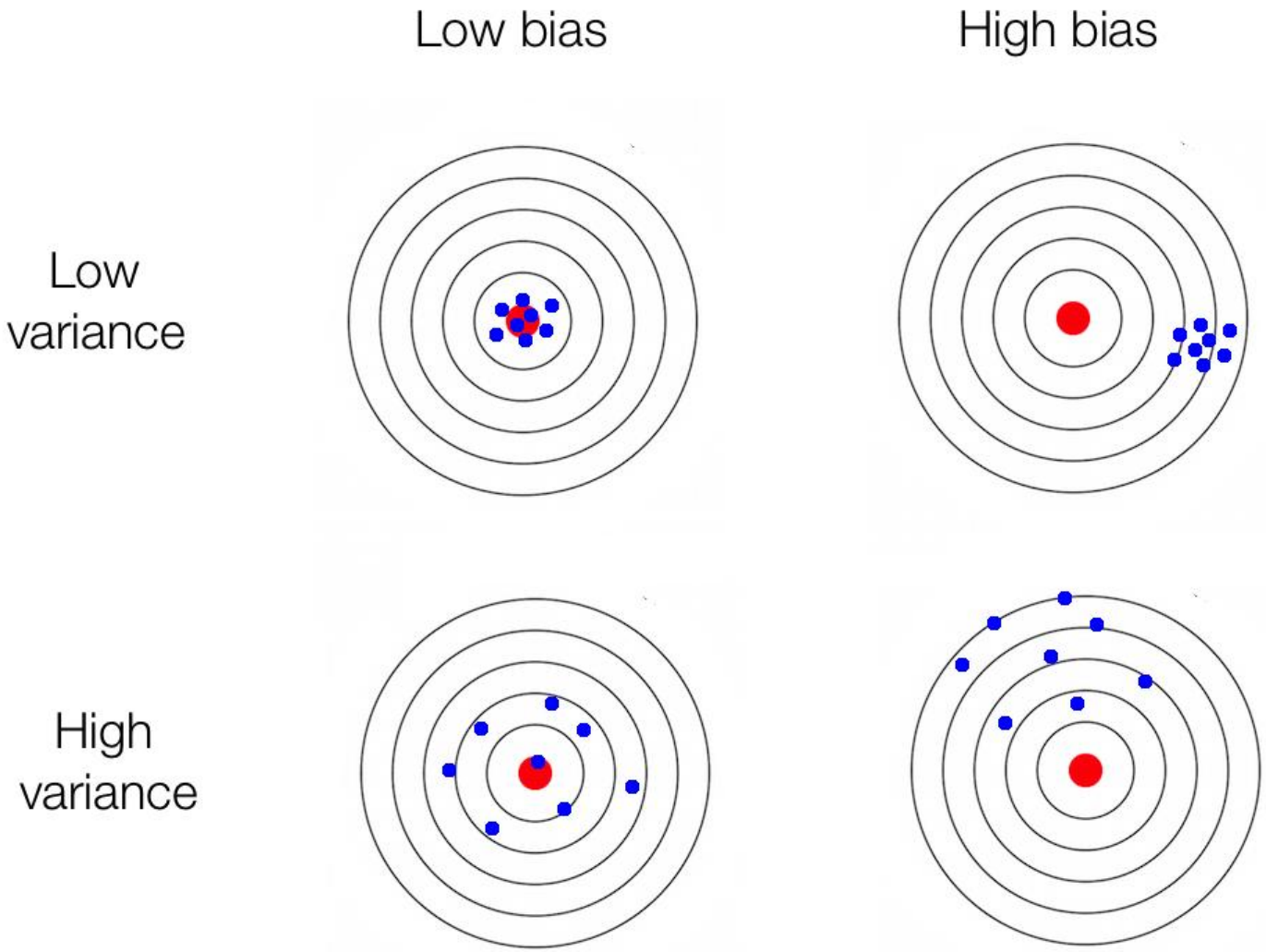


bias <> variance

Error due to bias (systematic errors, oversimplification)

Error due to variance (sensitivity due to flexibility, overcomplification)

Irreducible error

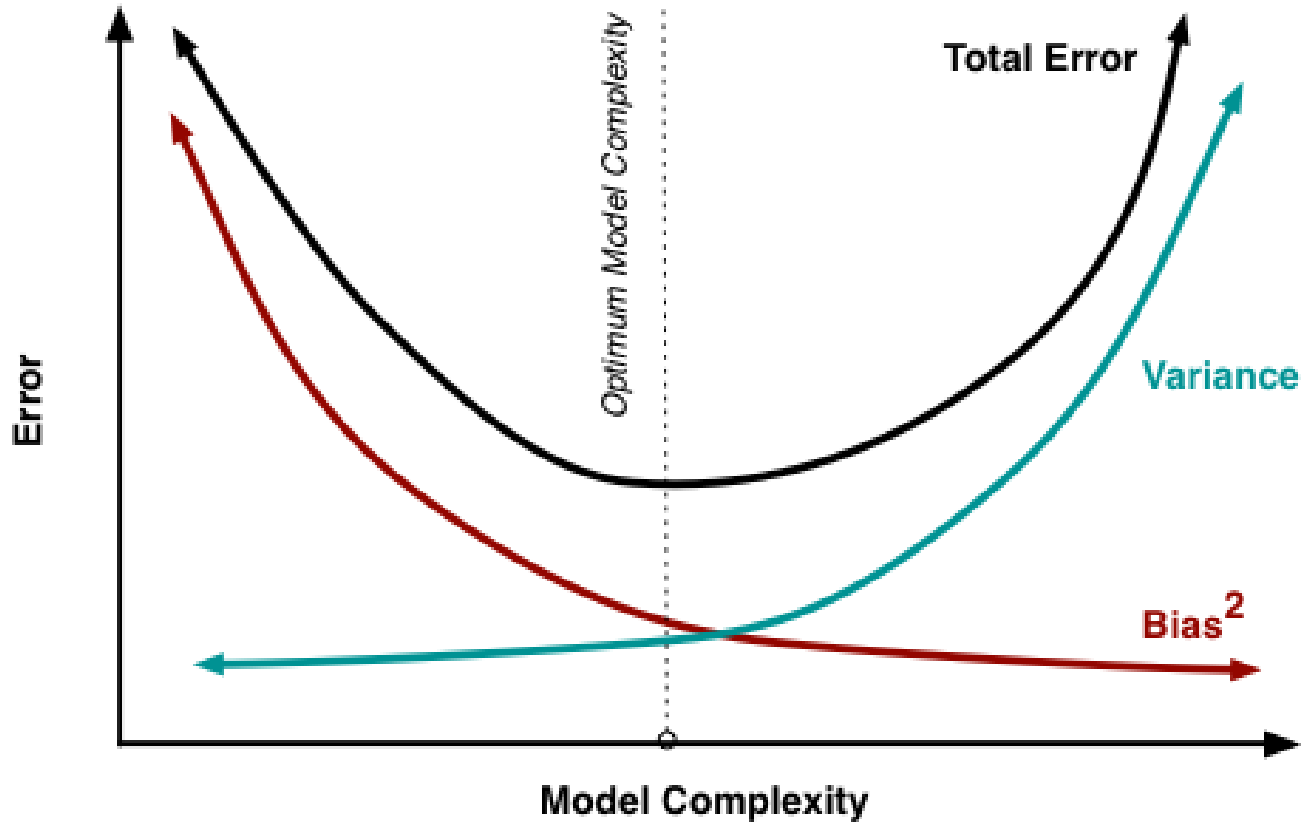


bias <> variance

Error due to bias

Error due to variance

Irreducible error



regularized linear regression

$$f(x,\theta) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \ldots + \theta_m x_m$$

cost function

$$J(\theta) = \frac{1}{2n} \sum_{i=1}^{n} (f(x^{(i)}, \theta) - y^{(i)})^2$$

$$J(\theta) = \frac{1}{2n} \sum_{i=1}^{n} (f(x^{(i)}, \theta) - y^{(i)})^2 + \lambda \sum_{j=1}^{m} \theta_j^2$$

regularized cost function

$$\theta_0 := \theta_0 - \alpha \frac{1}{n} \sum_{i=1}^{n} (f(x^{(i)}, \theta) - y^{(i)}) x_0^{(i)}$$

$$\theta_j := \theta_j - \alpha \frac{1}{n} \sum_{i=1}^{n} (f(x^{(i)}, \theta) - y^{(i)}) x_j^{(i)} - \frac{\lambda}{n} \theta_j$$

regularized logistic regression

$$f(x,\theta) = g(\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \ldots + \theta_m x_m)$$

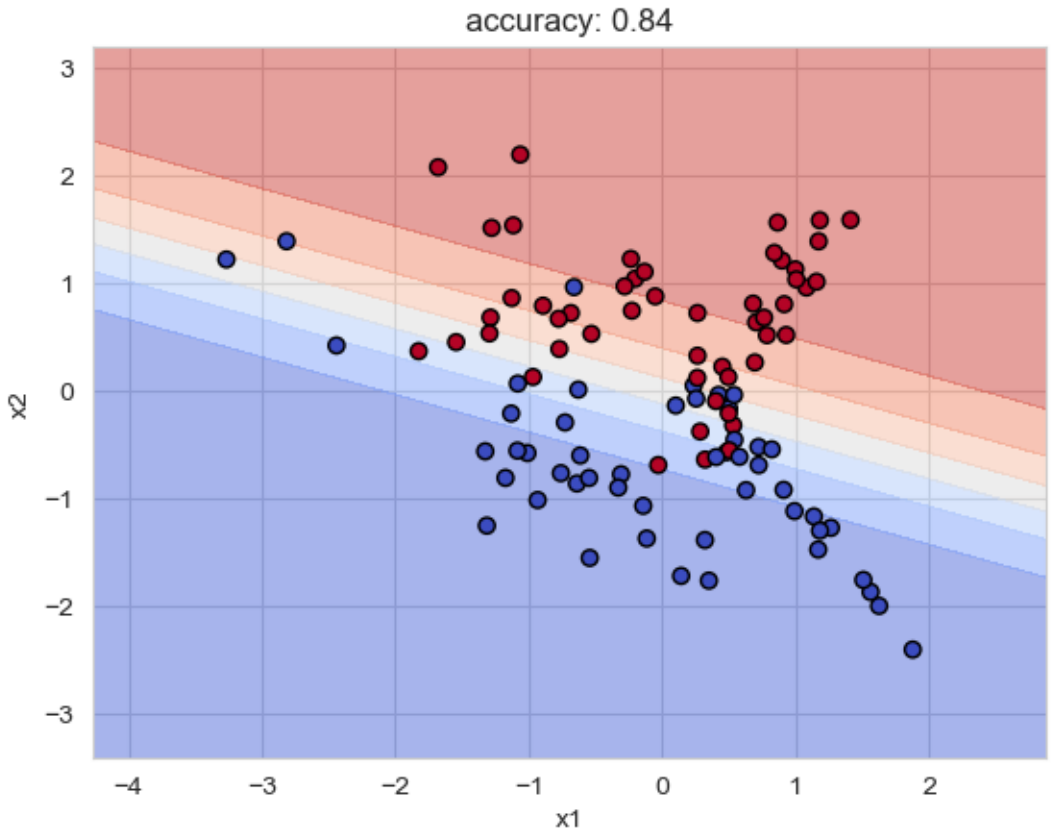
cost function

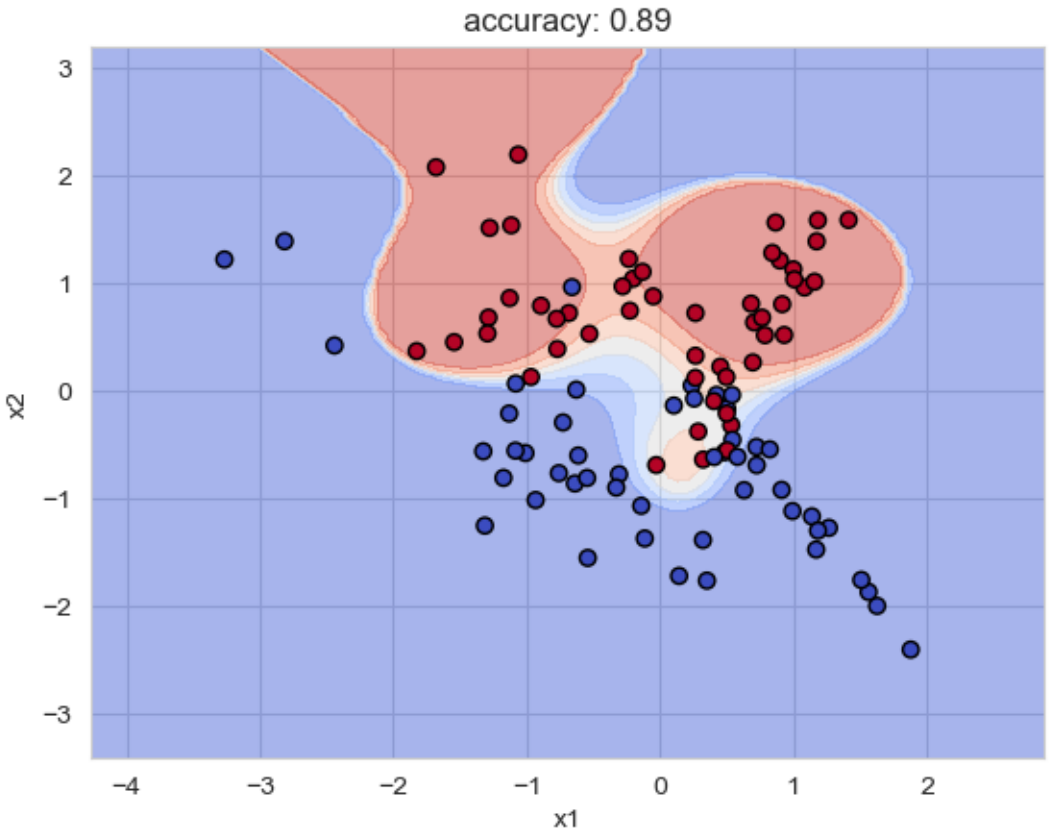
$$J(\theta) = -[\frac{1}{n}\sum_{i=1}^{n} y^{(i)} log(f(x^{(i)},\theta)) + (1 - y^{(i)}) log(1 - f(x^{(i)},\theta))]$$

$$J(\theta) = -[\frac{1}{n}\sum_{i=1}^{n} y^{(i)} log(f(x^{(i)},\theta)) + (1 - y^{(i)}) log(1 - f(x^{(i)},\theta))] + \frac{\lambda}{2m}\sum_{j=1}^{m} \theta_j^2$$

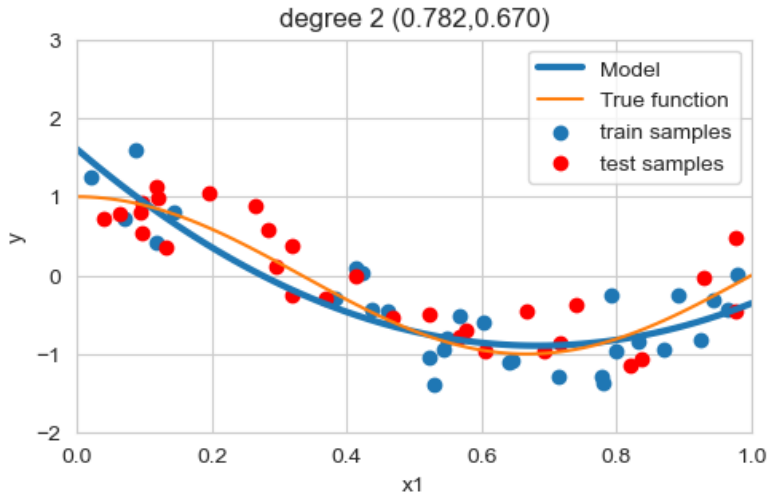
regularized cost function

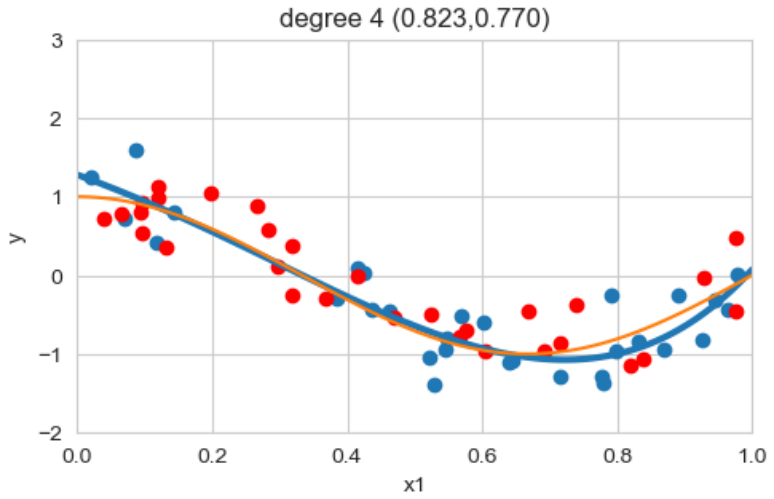
model regularization

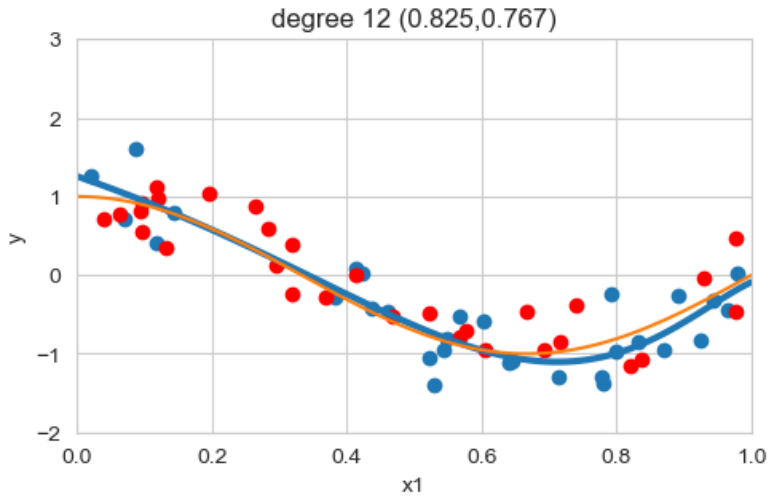


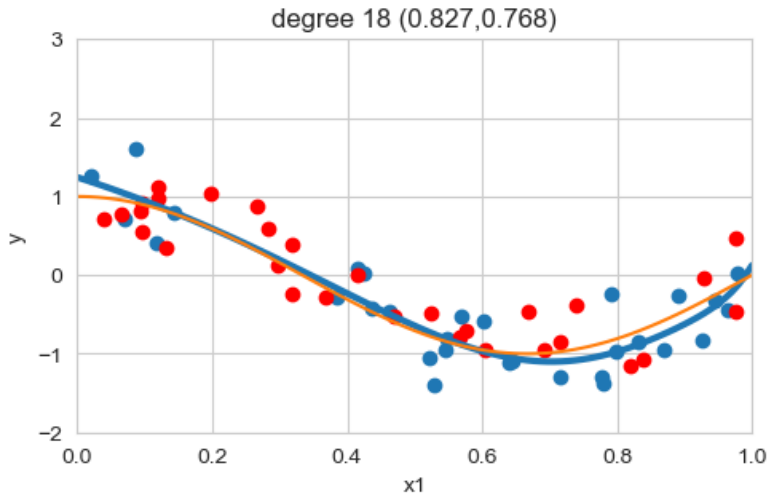


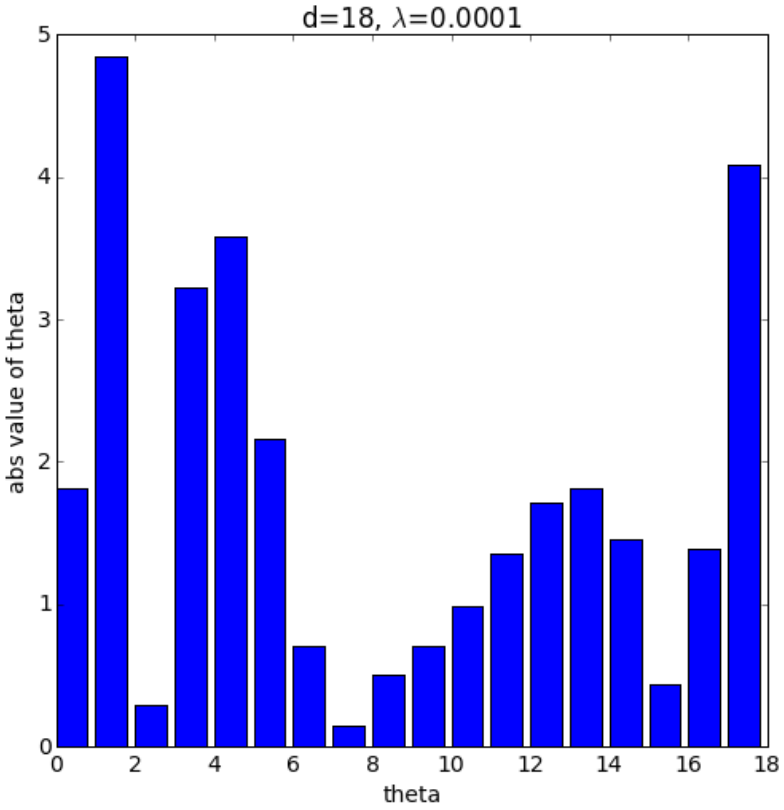
regularized linear regression



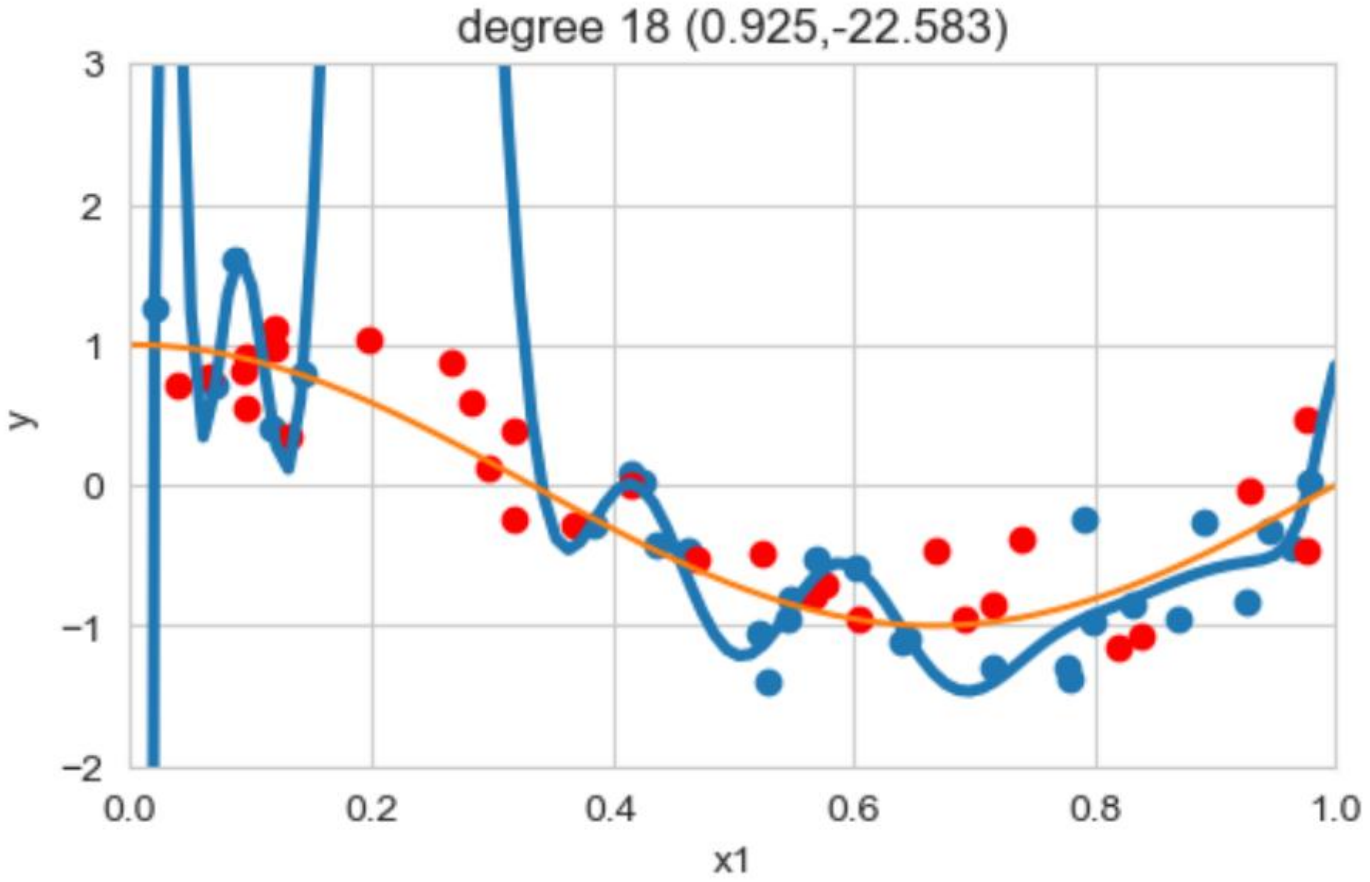


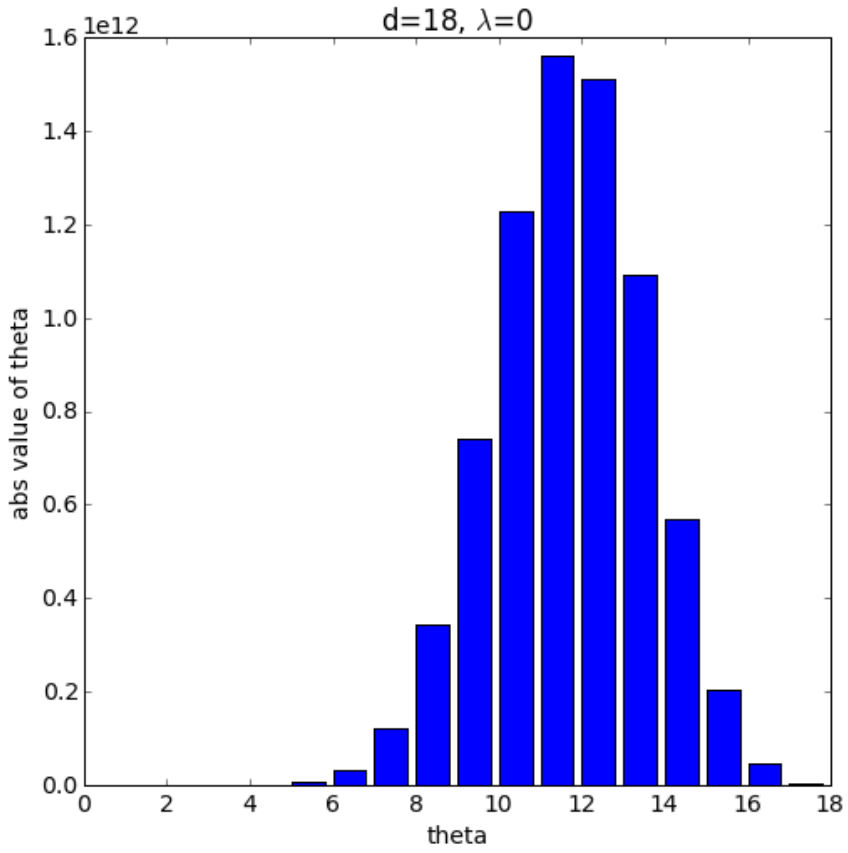






regularized linear regression





regularized linear regression

